

Sean Coath

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EDUCATION

University of Notre Dame

Notre Dame, IN | May 2029

Bachelor of Arts 4.00 / 4.00 (Dean's List)

Degrees/Programs: Computer Science Major (AI & Data Science Concentration), Philosophy Major (Philosophy of Science & Mathematics Concentration), Glynn Family Honors Program

Activities: Notre Dame Symphony Orchestra (Principal Tuba)

Notre Dame College Prep

Niles, IL | May 2025

High School Diploma 5.36 / 4.00 (Summa Cum Laude)

Honors: Valedictorian, National Merit Commended Scholar, AP Scholar with Distinction, IHSA Scholastic Award

COMPUTING EXPERIENCE

Data Inference Analytics & Learning (DIAL) Lab

Notre Dame, IN | Fall 2026

Incoming Research Assistant lucyinstitute.nd.edu/centers-and-labs/data-inference-analytics-and-learning-dial-lab/

- Will begin research in developing the RISE (Responsible, Inclusive, Safe, and Empowering) AI platform

University of Notre Dame Center for Research Computing

Notre Dame, IN | Summer 2024

Research Computing Scholar (Pre-College Program)

- Designed technical architecture & business proposal for an AI-driven college recommendation service
- Evaluated high-performance computing & AWS cloud infrastructure for scalable ML model training
- Collaborated with the ND Center for Research Computing & Argonne National Laboratory on data-intensive scientific modeling, distributed clusters, & enterprise-level systems
- Researched ethical constraints of student data privacy when analyzing open-source educational datasets

INDEPENDENT PROJECTS

Jeopardy!-Style Quiz Game | Python, Pygame, gTTS

- Leveraged Pygame for graphics, sound effects, AI-generated voice playback, & procedural animation
- Scraped ~500,000 clues & restructured into a JSON dataset for efficient filtration based on metadata
- Implemented nested data architecture to manage clue-state persistence for stat tracking & rules verification
- Designed a coordinate-mapping system to scale text, window, & assets for resolution-agnostic compatibility
- Engineered a reusable UI library to abstract complex asset management

SKILLS

Computer Science Coursework: Fundamentals of Computing, Discrete Mathematics, Calculus III

Technical Proficiencies: Python, C, HTML, Linux (basic)

Foreign Language: Intermediate Spanish (AP Spanish Language & Culture Score: 5)

LEADERSHIP & ACTIVITIES

Student Council & National Honor Society

Niles, IL | Aug 2022 – May 2025

Treasurer, Historian, Peer Tutoring Chair

- Elected by classmates to kickstart peer tutoring program serving ~400 underclassmen in all subjects
- Organized 15+ student events & fundraisers each school year while representing student opinions

Varsity Academic Bowl

Niles, IL | Aug 2021 – May 2025

Team Captain, Top Scorer

- Led practices & authored original study materials, contributing to multiple individual & team awards
- Coordinated volunteers & wrote questions for local middle school tournaments

Marching & Symphonic Band

Niles, IL | Aug 2021 – May 2025

First Chair Tuba, Brass Section Leader

- Mentored 5 new musicians & conducted rehearsals for 20+ school & community events each year
- Selected for ILMEA Senior Band 2x & received the highest award for musicianship & leadership

PROFESSIONAL EXPERIENCE

Self-Employed

Skokie, IL | Summer 2023 – Summer 2025

Landscape Maintenance Provider

- Maintained accurate records of service duration & project completion to ensure client satisfaction
- Managed consistent work schedule & high standards without direct supervision, resulting in 100% retention
- Communicated professionally with clients regarding project expectations & scheduling

SERVICE

Computer Science Mentor / Tutor, CovEducation

Chicago, IL | 2025 – present